Read a Text File in C++

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//Can I enter a full name separated by a space to
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3. Read a file

   Pro: can read a batch of data

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   4.1 Example of data.txt, where data are separated by a space or tab key

```
Name Test1 Test2 Test3
Ann 96 23 35
Bob 78 85 65
Carol 85 76 98
Deb 92 71 82
```

Class Encapsulates Data and Operations on Data

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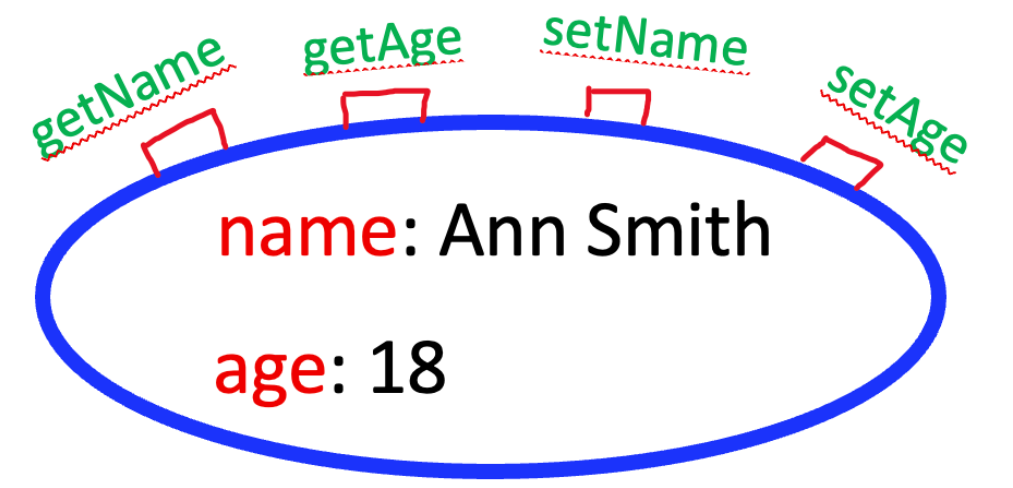
   2.2 setter (modifier): changeMeat, changeVeg, changeBread

Person class

Assume there is a class Person.



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   Example: declare ann as an object of Person class, representing a person named "Ann Smith" whose age is 18.

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Person ann("Ann Smith", 18);
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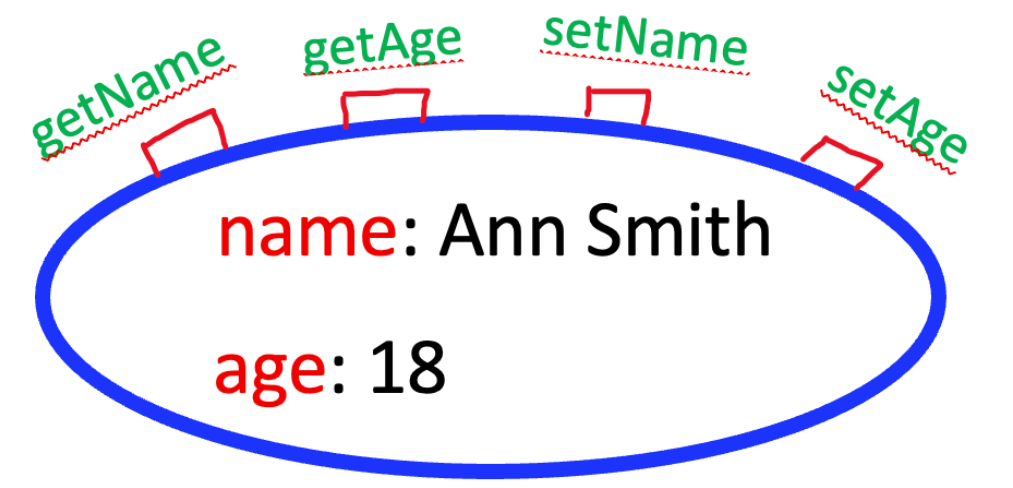

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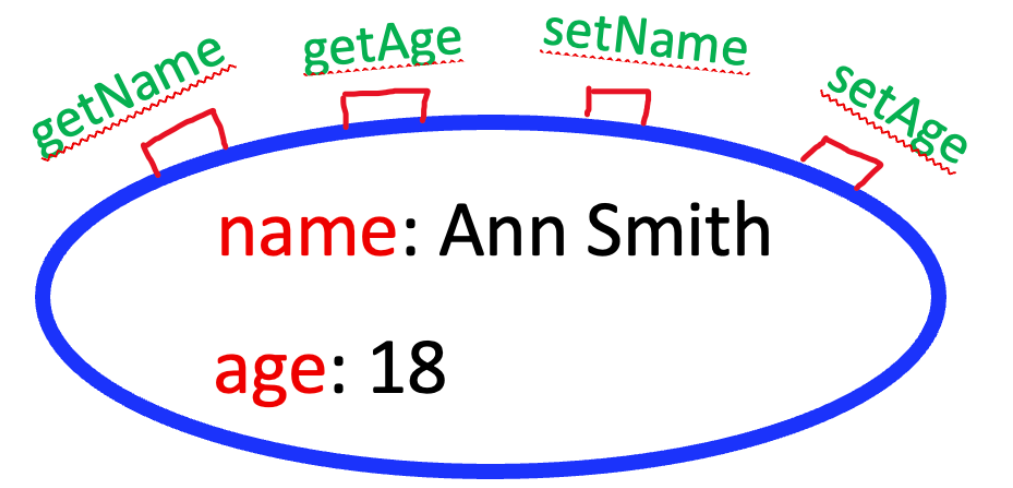

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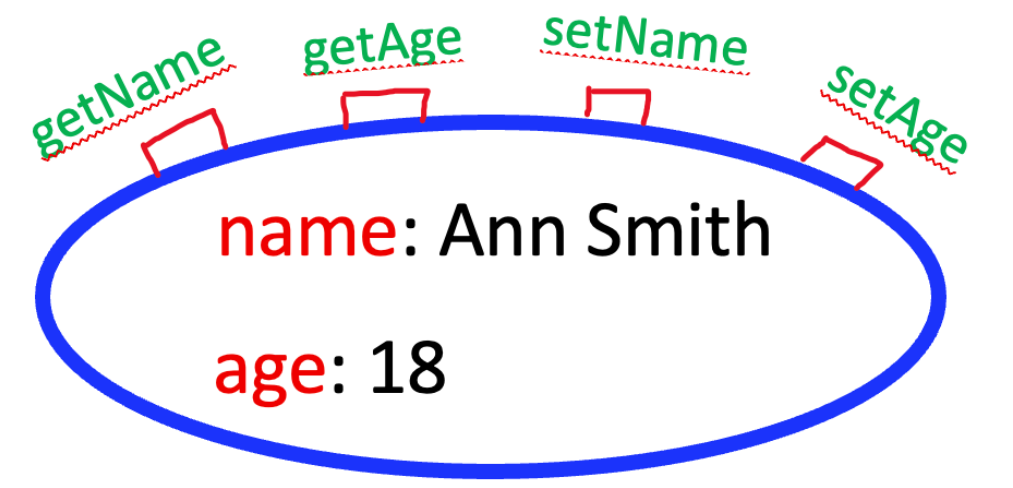

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   Data members of Person class are
   name and age.

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std::string name = ann.getName();
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```

   1.2 getAge: get the age of the current object,

```
int age = ann.getAge();
//ann.getAge() returns the age of Ann,//which
    is 18
```

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ann.changeName("Ann Smith Wood");
//change the name of ann to "Ann Smith Wood".
```

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```
ann.changeName("Ann Smith Wood");
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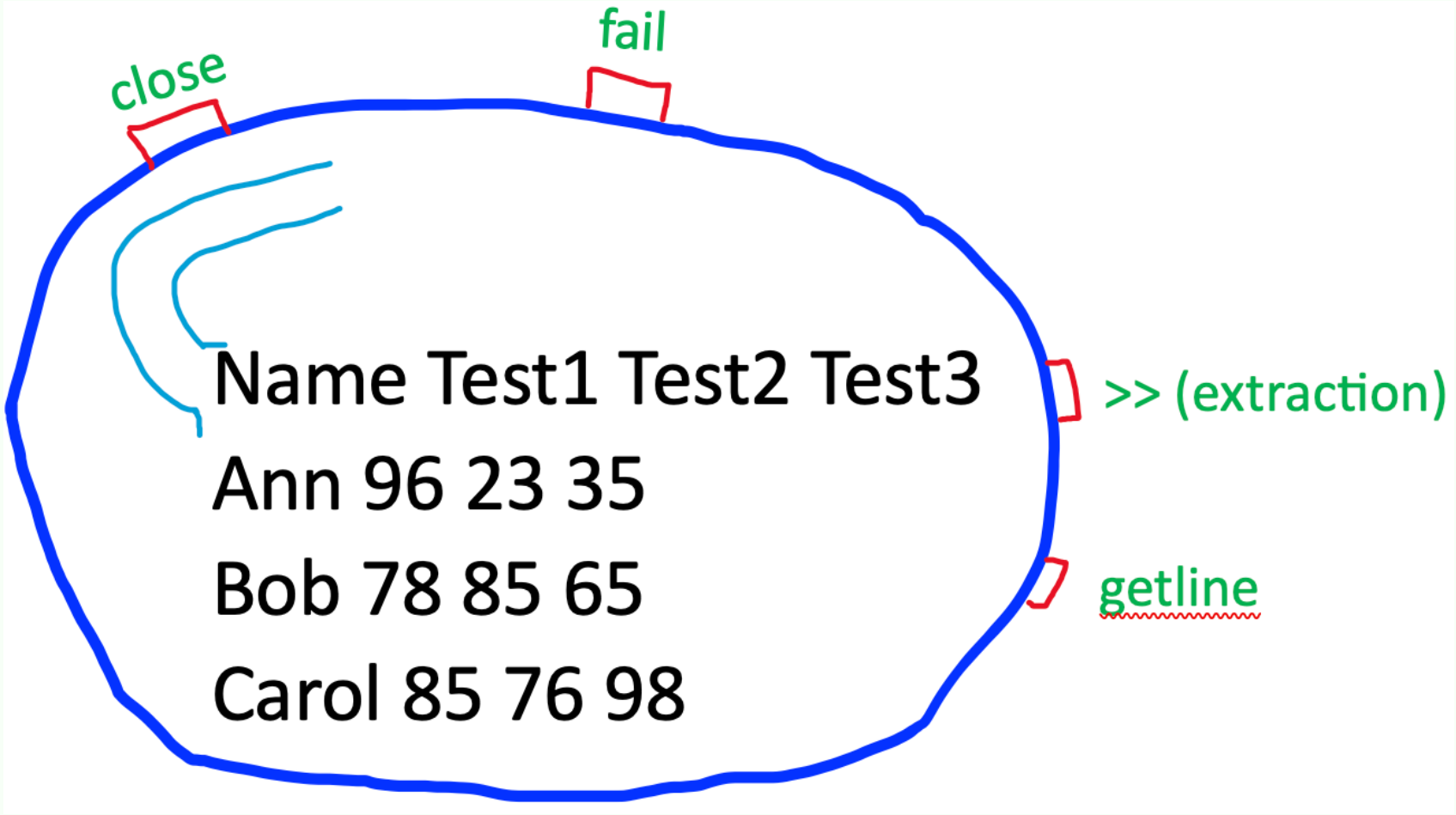
```

   1.2 changeAge: change the age of the current object

```
ann.changeAge(26);
//changes the age of ann to be 26
```

ifstream (input file stream) is a class

data and operations on those data



Use ifstream

1. Instantiate an `ifstream` object, use a valid variable name. For example, fin.

```
st::ifstream fin("data.txt");
//fin is an std::ifstream object
//that is connected to file "data.txt"
```

Warning: cannot write as

```
std::ifstream fin = "data.txt"; //WRONG
```

Reason:
`ifstream` is not a primitive data type.
example of a primitive data type is int

```
int count = 3;
```

instantiate a string object

string is an exception. A string object can be instantiated as one of the following approaches.

- Declare and initialize as if a string object were a primitive type.

```
std::string greeting = "Hello";
//C-style initialization
int i = 5;
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Reason: string is widely used in C and C++.

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- Declare and initialize as if a string object were a primitive type.

```
std::string greeting = "Hello";
//C-style initialization
int i = 5;
```

- Instantiate a string object using string constructor.

```
std::string greeting("Hello");
//instantiated greeting as a std::string object
```

Reason: string is widely used in C and C++.

Steps to Read a Text File

- Include necessary libraries.

```
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#include <climits> //INT_MAX
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std::ifstream fin("data.txt");
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- Test whether the file can be opened or not. That is, whether the ifstream object can properly connect to the file.

```
if (fin.fail()) {
   std::cerr << "file cannot be opened" << std::
       endl;
   exit(1);
}
```

Steps to Read a File: II

- Read the file contents, use ifstream object like it is a cin object. The difference is cin reads from keyboard, and ifstream object reads from a text file.

```
//Optional: skip explanation lines if they exist.
std::string line;
getline(fin, line);

//TODO: declare variables for columns.
//Suppose the names are variable1 and variable2.
while (fin >> variable1 >> variable2) {
    //TODO: process variables
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- Close the ifstream object when the file is finished reading.

```
fin.close();
```

Template to Read a Text File

```
//TODO: include proper libraries
int main() {
    ifstream fin("data.txt"); //file name can be input from console

    if (fin.fail()) {
        std::cerr << "file cannot be opened" << std::endl;
        exit(1); //exit with error 1 to operating system
    }

    //TODO: read file ...

    fin.close();
    return 0;
}
```

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- However, getline() can also be used with a custom delimiter. Given the following CSV file `scores.csv`, what is the delimiter?

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```

- Read from `streamObject`, stop right before `','` and put contents to `stringVariable`.

```
getline(streamObject, stringVariable, ',');
```

Read a CSV File: II

Exercise: Read `scores.csv` with the following contents, find out the minimum grade of Test 1.

```
Name,Test1,Test2,Test3
Ann Smith,96,23,35
Bob Taylor,78,85,65
Carol Miller,85,76,98
```

What are the key steps to finish this task?

Hints to Find the Minimum Grade of Test 1

- How to read the data?

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Hints to Find the Minimum Grade of Test 1

- How to read the data?

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  - If we have data for the first column – name in this example – do we have data for the rest columns?

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- `getline(streamObject, stringVariable, ',');` only save the data in a string variable.
- How to convert a string to an integer? Use `stoi` function.

```
int grade = std::stoi("123"); //grade is 123
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  - If we have data for the first column – name in this example – do we have data for the rest columns?
- `getline(streamObject, stringVariable, ',');` only save the data in a string variable.
- How to convert a string to an integer? Use `stoi` function.

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- How to initialize the minimum value?